

[Logo: Hackles]

Concepts Lab

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-in-inked, in the order the lessons take them.

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OCTOBER 2026

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Hackles • Cybersecurity Concepts Lab for Teachers

Demo 2: Mystery Fingerprints (Why Websites Store Fingerprints, Not Passwords)

Trust your hackles. Pause • Check • Tell

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE

Big idea	A hash is a one-way "fingerprint" of data. A site can check your password without keeping it, and a stolen list of fingerprints is much less harmful than a stolen list of passwords.
Grade bands	K-2 (the "smoothie machine" idea only), 3-5 (full game), 6-8 (full game + collisions), 9-12 (full game + salts, slow hashes, integrity checks)
CSTA-style concepts	Networks and the Internet (cybersecurity), Data and Analysis, Algorithms and Programming (functions as steps) (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Paper, pencil, a printed word list. Optional calculator. No devices.
Teachers do it first	Yes. Part A takes about 20 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Pair up. Use the recipe in Section 3 on the word "dog": start at 7; for each letter multiply by 3, add its number (a=1 ... z=26), keep the last two digits. Your partner checks you both get the same two-digit "fingerprint". Can either of you get from the fingerprint back to the word without just trying words?

01 The big idea

IF A WEBSITE KEPT your password in a file, anyone who stole the file would have everyone's passwords. So good systems keep only a **fingerprint** of the password. The fingerprint is made by a recipe called a **hash function**. When you log in, the site makes a fingerprint of what you typed and checks whether it matches the saved one. The site can say "yes, that's the right password" without ever remembering it.

The key properties:

1. **Same input, same fingerprint, every time.**
2. **Easy to make, practically impossible to run backward.** You cannot "un-blend" a smoothie.
3. **A tiny change gives a totally different fingerprint.** (cat vs car)
4. **Fixed size.** A word or a movie gives the same length of fingerprint.

And one big honest caveat for teachers: **hashing does not make a weak password strong**. If your password is on a list of common passwords, an attacker can fingerprint every word on the list and see which one matches. That is why Demo 1 (long, random) and this demo (store fingerprints, add salt, use slow recipes) work together.

02 What a teacher should know (plain language)

2.1 HASHING, ENCRYPTION AND ENCODING ARE THREE DIFFERENT THINGS

TOOL	REVERSIBLE?	NEEDS A KEY?	PURPOSE
Encoding (for example Base64)	Yes, by anyone	No	Change format. Not secret at all.
Encryption (Demo 4)	Yes, with the key	Yes	Keep a message secret but still readable later by the right person.
Hashing	No (by design)	No (not a secret recipe)	Make a fingerprint: check a password, check a file has not changed.

NEWS STORIES OFTEN SAY "the passwords were encrypted" when the correct word is usually "hashed." The difference matters: encrypted data can be

decrypted by anyone who has the key; hashed passwords are never "decrypted," only guessed.

2.2 REAL HASH FUNCTIONS

- **SHA-256** makes a 256-bit fingerprint (64 hexadecimal characters). The number of possible fingerprints is about 1.16×10^{77} [RECOMPUTED 2026-10-05: $2^{256} = 1.158 \times 10^{77}$, a 78-digit number], so accidental matches (collisions) are not a practical worry.
- Real example of "tiny change, totally different fingerprint" (SHA-256, computed with a standard tool):
 - "cat" starts 77af778b...
 - "car" starts 2b2961a4...
 - "Cat" starts 48735c4f...
- **Old hashes are retired.** MD5 and SHA-1 have known collision weaknesses and should not be used for security [VERIFIED 2026-10-05: NIST says SHA-1 should be phased out by December 31, 2030 in favor of SHA-2 and SHA-3, and shattered.io documents the first SHA-1 collision in February 2017].
- **Password storage should use deliberately slow, salted hashing:** algorithms such as Argon2, scrypt, bcrypt or PBKDF2 (the OWASP Password Storage Cheat Sheet is a plain reference) [VERIFIED 2026-10-05: OWASP lists Argon2id first (minimum 19 MiB memory, 2 iterations, 1 degree of parallelism), then scrypt, bcrypt for legacy systems (work factor 10 or more) and PBKDF2 where FIPS compliance is needed (600,000 or more iterations with HMAC-SHA-256)]. Slow is a feature: it makes billions-per-second guessing (Demo 1) cost much more. Plain SHA-256 is fast, so it suits checksums but is not by itself a good password-storage method.

2.3 SALT, IN ONE PICTURE

A **salt** is a random extra bit of text added to each person's password before hashing, stored next to the fingerprint (it is not secret). Without salt, two people with the same password get the same fingerprint, and an attacker can prepare one big list of fingerprints for common words ahead of time (often called a "lookup table", or in a famous variant a "rainbow table"). With a unique salt per user, the prepared list is useless and every user's password has to be attacked separately. Salt does not make a weak password strong; it stops mass shortcuts.

2.4 OTHER EVERYDAY USES OF HASHES



- **File integrity.** Download pages sometimes publish a checksum. Your computer makes a fingerprint of the download; if it matches, the file did not change on the way [a checksum posted on the same compromised site does not prove authenticity; digital signatures do; standard security reasoning].
- **Breach checks.** Services can tell you whether your password appears in known breaches without learning your password, by comparing partial fingerprints (the approach used by "Have I Been Pwned" Pwned Passwords: only the first 5 characters of the password's SHA-1 fingerprint are sent, and the service returns all matching endings) [VERIFIED 2026-10-05: haveibeenpwned.com/API/v3].

2.5 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"If it's hashed, it's safe."	Only as safe as the password is hard to guess. Weak passwords fall to guessing.
"Hashing and encryption are the same."	Encryption is reversible with a key; hashing is one-way.
"A hash tells you nothing about the input."	It reveals when two inputs are identical, which is exactly why unsalted lookup tables work.
"Hash collisions mean it's broken."	Collisions must exist (infinite inputs, finite outputs). Good hashes make finding one infeasible.
"Salt must be secret."	No. It just needs to be unique and random.
"Websites should email you your password."	A site that can email your password is storing it in readable form. Red flag.

03 Materials (zero to low cost)

- The **Fingerprint Recipe card** (below), one per pair.
- The **Class Word List** (16 words, below), one printed per pair.
- Index cards or strips of paper for "secret words" (students choose from the Class Word List so we never use real passwords).
- A large "Ledger" sheet of chart paper or whiteboard with three columns: Name, Fingerprint, Salt.
- Pencil, calculator (optional), a cup to hold folded secret slips.

The Toy Fingerprint Recipe (for grades 3 and up)

1. Turn each letter into a number: a=1, b=2, c=3 ... z=26.
2. Start with the number 7.
3. For each letter in order: **multiply your current number by 3, add the letter's number, then keep only the last two digits** (divide by 100 and keep the remainder).

4. The last two-digit number you reach is the **fingerprint**.

Worked example, cat: start 7. $c=3: 7 \times 3 + 3 = 24$. $a=1: 24 \times 3 + 1 = 73$. $t=20: 73 \times 3 + 20 = 239$, keep 39. **Fingerprint of cat = 39.**

Check yourself: dog = 77.

Class Word List (16 words; printed on the student handout): cat, car, cot, dog, pizza, moon, sun, hat, tiger, robot, panda, cake, star, bike, fish, frog. The fingerprints are on the teacher answer page. Two pairs of words collide by accident, because there are only 100 possible fingerprints. That is the toy's weakness and a perfect teaching moment (Part B).

Toy salt rule. Put the salt letter in front of the word before hashing. Same word, different salt letter, different fingerprint (examples on the teacher answer page).

Teacher note: this toy is **not secure**. It exists to make the ideas physical. Say so clearly: "Real fingerprints are 64 characters long, not two digits."

04 Part A: Teachers do it first (about 20 minutes)

1. **Learn the recipe (5 min).** Compute "cat", "dog" and "moon" by hand and check the answer page. You cannot work backwards from 77 to "dog" by arithmetic. You can only try words.
2. **Register and log in (6 min).** Each teacher picks a secret word from the list and writes name plus fingerprint on a shared Ledger. One teacher announces "I'm Pat, my password is cat"; the "booth keeper" fingerprints it and compares. Try a wrong word.
3. **The Stolen Ledger (5 min).** Imagine the ledger was stolen. Try to find a colleague's word by fingerprinting every word on the list. With 16 words it takes minutes; with long random passphrases it never ends (Demo 1).
4. **Salt it (4 min).** Re-register with a salt letter drawn from a bag. A prepared word-to-fingerprint cheat sheet no longer works, and two people with the same word now differ.

05 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook: the teacher reads the script. Introduce the Fingerprint Recipe (or the smoothie machine for grades K-2).
2-5	Teach by example: compute "cat" together on the board.
5-9	Register and login round: students pick a word, compute, post fingerprint on the Ledger. Test two logins.
9-12	The Stolen Ledger: "A burglar took the ledger!" Teams race to find who has which word. Debrief: weak, common words fall fast.
12-15	Salt: apply a one-letter salt and see the fingerprints change. Exit slip.

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-5	Hook and smoothie warm-up (below).
5-15	Learn and practice the recipe; students compute five words; check and correct mistakes.
15-25	Register, login test, wrong-password test. Discuss: "How does the booth know you're right without a copy of your word?"
25-33	The Stolen Ledger race: teams compete for speed in decoding the ledger. Debrief on weak passwords and list size.
33-40	Collisions: show cot and star both give 81. Why does this happen with 100 fingerprints? How many would we need? Introduce real SHA-256 examples (cat vs car vs Cat) on the board. Salt round.
40-45	Exit ticket and handout.

Smoothie warm-up (all bands, 3 minutes). Show three index cards labeled "strawberry + banana", "strawberry + mango", "strawberry + banana" (again). Say: "A smoothie machine (the hash) turns a recipe into a color: pink-banana, orange, pink-banana. Same recipe, same color. Can anyone look at a pink smoothie and say exactly what recipe it was?" Discuss: "No, but if I tell you the choices, you can try each."

06 Teacher script

Hook (2 minutes). "I have a secret word and I'm going to prove I know it without ever telling you or writing it down. Here's the trick: I'm going to use a recipe that turns any word into a two-digit number, like a fingerprint. If I give you the fingerprint, you can check that my word is right, but you can't easily work out the word. Today we are the website's computer, and we find out why websites store fingerprints, not passwords."

After the Stolen Ledger. "The burglar had a list of fingerprints. They could try each word on a list and compare. That is exactly what happens in real life when a company is breached, only with billions of tries. Which of your classmates' secret words got found fastest? The common ones. That's why Demo 1's long random passphrases matter, and why companies use salts."

On collisions. "Our toy has only 100 fingerprints, so different words can share one. A real SHA-256 fingerprint has about 10^{77} possibilities (a number with 78 digits), compared with a commonly cited estimate of about 10^{80} atoms in the observable universe [commonly cited estimate; not checked against a primary source], so accidents basically don't happen. That's why real ones are 64 characters long."

Close. "A website that can email you your password is storing it in the open. A good website stores only fingerprints, with salt, made by a slow recipe, and checks yours when you log in. You can help by choosing a long, random passphrase and a different one for each site. If you ever get a message that your account was in a breach, it's not your fault; change that password and tell a trusted adult."

07 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	The smoothie machine. The same fruit makes the same color. You cannot un-blend it. Play "guess the fruits": the teacher shows only the smoothie color card and the class guesses which recipe from three choices.
3-5	The fingerprint booth. Teacher or helper runs the recipe; students post fingerprints; the "burglar" gets the ledger and tries words. Add salt as a "secret sprinkle" that changes the color for each person.
6-8	Hash, collision, salt. Compute by hand, find collisions (cot and star), discuss why more fingerprints means fewer collisions, and why salting matters. Introduce "avalanche": one letter changes everything in real hashes.
9-12	Storage design and attack economics. Slow hashes (Argon2, bcrypt), salts and peppers (a pepper is a secret value kept outside the password database; OWASP's Password Storage Cheat Sheet has a "Peppering" section) [VERIFIED 2026-10-05: cheatsheetseries.owasp.org], breach response, checksum vs signature, why "we encrypt your passwords" is a weaker claim than "we hash them with a slow, salted algorithm".

08 Discussion questions

1. How can the booth check your password without keeping it?
2. Why did the burglar find some words quickly and others slowly?
3. If two people pick the same word, why should their fingerprints still be different?
4. (9-12) Why do systems deliberately use *slow* hashing for passwords but *fast* hashing for file checksums?



09 Level-up challenge

- **Find a collision.** Find two *different* three-letter words with the same fingerprint, then explain why finding one for SHA-256 is thought infeasible while ours takes minutes (a bigger fingerprint). This teaches nothing about attacking a real system.
- **Spot the policy.** A pretend privacy policy says "we store your password so we can email it if you forget." Students write a one-sentence critique and say what the company should do.

10 Safety and ethics notes (defense only)

- The "burglar" role is a **thinking role** with a list on paper, never a real attack on real accounts. Do not use real passwords at any time; all secret words come from the Class Word List.
- Do not show or teach cracking tools or run them against any real hash. Students learn why guessing works so they choose better passwords, not how to attack databases. If someone says the toy "proves hashing is breakable," clarify that it shows weak secrets and tiny spaces are breakable.
- If a student asks "can I crack someone's password?" redirect: "Only with permission and for defense. Let's talk about ethical hacking in Demo 14."
- Keep the "burglar" friendly (a silly cartoon), not a scary stranger. Hackles up? Pause. Check. Tell.

11 Assessment idea

- **Exit ticket.** "Why can a website check your password without keeping it? Use the word 'fingerprint'." Rubric: 3 = same word gives same fingerprint, can't reverse, site compares; 2 = says "it's scrambled"; 1 = says "it's encrypted so no one can see".
- **Performance task (6-12).** Give a short ledger (names, fingerprints, salt) and a word list; students recover the secret words, then write one paragraph explaining why salt slowed them down and what the company should do about weak passwords.

12 Sources and [VERIFY] list

- NIST FIPS 180-4 (SHA-2 family; final August 2015) [VERIFIED 2026-10-05: csrc.nist.gov].
- OWASP Password Storage Cheat Sheet (Argon2id, bcrypt, scrypt, PBKDF2 guidance) [VERIFIED 2026-10-05].
- NIST SP 800-63B-4 (final August 2025), password verification guidance [VERIFIED in Round 1: pages.nist.gov/800-63-4].
- MD5 and SHA-1 collision weaknesses (CWI/Google 2017 "SHattered" for SHA-1 [VERIFIED 2026-10-05: shattered.io]; earlier results for MD5 [not checked]).
- Troy Hunt's "Have I Been Pwned" k-anonymity range search for password checks [VERIFIED 2026-10-05: haveibeenpwned.com/API/v3; the project's author attribution was not re-checked].
- Inspiration for unplugged cryptography activities: CS Unplugged (University of Canterbury, New Zealand; content shared under CC BY-SA 4.0 [VERIFIED 2026-10-05: csunplugged.org]), cryptography and information-hiding activities. This demo is an original design, not a copy; check the specific activity name and license before adapting any CS Unplugged content directly.

Student Handout: Mystery Fingerprints

Name: _____

The big idea

A **WEBSITE KEEPS A fingerprint** of your password, not the password. When you log in, it makes a fingerprint of what you type and checks for a match.

The recipe (Toy Fingerprint Machine)

1. Letters become numbers: a=1, b=2, c=3 ... z=26.
2. Start at 7.
3. For each letter: **multiply by 3, add the letter number, keep the last two digits.**

Example: **cat** gives 7, then 24, then 73, then 39. Fingerprint = **39**.

Class Word List: cat, car, cot, dog, pizza, moon, sun, hat, tiger, robot, panda, cake, star, bike, fish, frog

Your turn

Show your work in the margin.

WORD	FINGERPRINT
dog	_____ (check: 77)
moon	_____
fish	_____
bike	_____

The Stolen Ledger

YOUR TEACHER POSTS A ledger of names and fingerprints. Fingerprint every word on the Class Word List, then find who has which secret word:

FINGERPRINT	WHICH WORD?
53	_____
12	_____
43	_____
95	_____

2

Add salt!

SALT = A RANDOM extra letter in front of the word. Compute the fingerprint of **kcat**: ____ . Is it the same as **cat** (39)? ____

Think about it

1. Can you work backward from 53 to the word using arithmetic? ____
2. Two different words on the list give the same fingerprint (81). What are they? ____ and ____
3. A real fingerprint is 64 characters long. Why does a longer fingerprint mean fewer accidental twins?

Remember

- Hashing is one-way. Encryption is reversible with a key.
- Weak, common passwords are found quickly even if hashed. Make yours **long and random**.
- If your password is ever stolen or in a breach, it is **not your fault**. Change it and tell a trusted adult.

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 2 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Master fingerprints (toy recipe, verified):

WORD	FINGERPRINT	WORD	FINGERPRINT
cat	39	tiger	60
car	37	robot	47
cot	81	panda	63
dog	77	cake	95
pizza	53	star	81
moon	12	bike	40
sun	37	fish	75
hat	84	frog	43

Accidental twins: cot and star both give 81; car and sun both give 37. With only 100 possible fingerprints, collisions are common.

Salt examples: kcat = 14, mcat = 68, qcat = 76 (cat alone = 39).

Handout answers: moon 12, fish 75, bike 40. Ledger: 53 pizza, 12 moon, 43 frog, 95 cake. kcat = 14, not the same as cat (39). Think about it: 1 No, you can only try words; 2 cot and star; 3 more possible fingerprints means two words are much less likely to share one.

Real SHA-256 examples (computed with a standard tool): cat 77af778b..., car 2b2961a4..., Cat 48735c4f... One changed letter, a completely different fingerprint.